RULES AND REGULATIONS FOR THE WRITTEN EXAMS

AT THE FACULTY OF ENGINEERING AND NATURAL SCIENCES

Please read this document carefully, sign the bottom part, and include this form with your solution sheet.

- The exam duration (start-end times) will be written on your exam sheet or will be announced, you should adhere to these times.
- You cannot leave the examination room in the **first 30 minutes** of the exam.
- You are **NOT allowed** to take the exam once a student has left one of the exam rooms.
- Talking with another student during the exam for any reason (asking for an explanation, for a calculator or eraser or any other object) is **NOT allowed**.
- If you **GIVE** to or **RECEIVE** from a student any item (eraser, calculator, pen, etc.) during the exam, this will be taken as a sign of cheating.
- **TURN OFF** your **CELLPHONES AND SMART DEVICES**; and PLACE THEM **away from you** such that those **CANNOT BE SEEN** from outside.
- **HAVING a CELLPHONE or a SMART DEVICE** that is VISIBLE FROM OUTSIDE during the exam is **NOT ALLOWED**. Interacting with a cellphone in any way (EVEN IF IT IS TURNED OFF) will be taken as a sign of cheating.

In case of a sign of cheating, or if you do not obey the above rules, there will be a disciplinary action taken towards you based on the STUDENT DISCIPLINARY REGULATIONS of YÖK Items 2.5, 2.7, and 2.8.

I understand the above-mentioned rules and regulations, and I hereby accept them, and any consequences following my actions during the exam.

Student Full Name:

Student Number:

Course Code and Title:

AIN-2002 INTRODUCTION TO DATA SCIENCE

Signature:

Date:

JUNE 1st, 2024

INTRODUCTION TO DATA SCIENCE

FINAL EXAM

(This is a 100-minute exam)

Q.1	**Q.2**	**Q.3**	**Q.4**	**Q.5**	**TOTAL**

Q.1) (15) Data comes in various formats, impacting how it can be stored, analyzed, and utilized. Distinguish between the following data types by providing real-world examples for each:

Structured Data Definition: Clear and relevant example:
Unstructured Data Definition: Clear and relevant example:
Semi-structured Data Definition: Clear and relevant example:

Q.2) (25) A telecommunications company like Turk Telekom is experiencing a high customer churn rate (i.e., customers cancelling their service). To develop strategies for customer retention, they aim to predict which customers are at risk of churning. The company has collected customer data, including demographics, service plans, usage patterns, and past support interactions. Here is a sample dataset in CSV format based on the problem definition:

CUSTID	Gender	Age	ContractType	MonthlyCharges	TotalCharges	Service Usage	Support Calls	Churn
1	Female	45	One year	70.35	1200.50	High	2	No
2	Male	32	Month-to-month	56.75	450.20	Medium	5	Yes
3	Female	29	Two year	89.50	2400.75	Low	1	No
4	Male	52	One year	45.00	1100.00	Medium	3	No
5	Female	37	Month-to-month	99.99	150.99	High	7	Yes
6	Male	41	Two year	60.80	1800.40	Low	0	No
7	Female	34	One year	75.20	900.00	Medium	4	No
8	Male	25	Month-to-month	40.50	300.00	High	6	Yes
9	Female	55	Two year	55.00	1700.00	Low	1	No
10	Male	38	One year	85.75	1300.50	Medium	2	No
11	Female	28	Month-to-month	65.30	600.75	High	8	Yes
12	Male	46	Two year	70.00	2000.00	Low	0	No

Column Descriptions:
CUSTID: Unique identifier for each customer.
Gender: Gender of the customer.
Age: Age of the customer.
ContractType: Type of contract (Month-to-month, One year, Two year).
MonthlyCharges: The amount charged to the customer monthly.
TotalCharges: The total amount charged to the customer.
ServiceUsage: Overall usage level of the service (Low, Medium, High).
SupportCalls: Number of support calls made by the customer.
Churn: Indicates whether the customer has churned (Yes, No).

Churn, in the context of business and telecommunications, refers to the phenomenon where customers stop using a company's products or services and leave for a competitor or discontinue the service altogether. This is often measured as a churn rate, which is the percentage of customers who cancel or do not renew their subscriptions over a given period. In the dataset provided, the Churn column indicates whether a customer has churned or not:

- Yes: The customer has canceled their service and is no longer using the company's products.
- No: The customer has not canceled their service and is still using the company's products.

a) (10) Identify a suitable supervised learning algorithm for predicting customer churn in this scenario. Briefly explain why this algorithm is appropriate for this problem.

b) (10) Describe the key steps involved in building a customer churn prediction model using the chosen algorithm. Briefly explain what happens at each step.

c) (5) Once you have a trained model, how can the company use it to identify customers at risk of churning?

Q.3) (20) A credit card company is constantly on the lookout for fraudulent transactions. They receive a massive volume of transaction data daily, including details like purchase amount, location, merchant category, and time of purchase from physical POS (Point of Sale) devices. Physical POS is a payment device allowing a company to receive payments from all debit and credit cards. The company wants to develop a system to identify potentially fraudulent transactions in real time.

Here is a sample dataset in CSV format based on the problem definition. This dataset includes various features that could help in identifying fraudulent transactions, such as purchase amount, location, merchant category, and time of purchase:

```
TransactionID,CardholderID,PurchaseAmount,Location,MerchantCategory,TimeOfPurchase,Fraudulent
1,1001,50.25,New York,Groceries,2024-05-29 08:15:30,No
2,1002,4500.00,California,Electronics,2024-05-29 14:22:10,Yes
3,1003,75.50,Montana,Restaurants,2024-05-29 12:35:50,No
4,1004,200.75,New York,Clothing,2024-05-29 09:45:20,No
5,1005,5.00,Florida,Convenience Store,2024-05-29 07:05:15,No
6,1006,4500.00,Alabama,Jewelry,2024-05-29 16:50:45,Yes
7,1007,120.00,Illinois,Gas Stations,2024-05-29 11:30:05,No
8,1008,35.40,New York,Entertainment,2024-05-29 20:15:25,No
9,1002,3000.00,Nevada,Luxury Goods,2024-05-29 12:55:55,Yes
10,1010,60.00,Missouri,Groceries,2024-05-29 13:20:35,No
11,1011,85.75, Washington,Restaurants,2024-05-29 10:40:00,No
12,1012,40.50,Texas,Pharmacy,2024-05-29 08:25:10,No
13,1006,5000.00,Washington,Electronics,2024-05-29 16:45:30,Yes
14,1014,15.75,Oregon,Convenience Store,2024-05-29 06:45:20,No
15,1015,80.25,Minnesota,Clothing,2024-05-29 14:30:00,No
16,1019,225.00,New York,Jewelry,2024-05-29 17:05:10,Yes
17,1017,95.60,Nevada,Entertainment,2024-05-29 12:50:45,No
18,1018,60.00,San Francisco,Gas Stations,2024-05-29 09:15:25,No
19,1019,3500.00,Arizona,Luxury Goods,2024-05-29 17:06:55,Yes
20,1020,25.00,Rhode Island,Groceries,2024-05-29 11:20:35,No
21,1021,45.75,Indiana,Pharmacy,2024-05-29 08:30:10,No
22,1022,150.00,Wyoming,Electronics,2024-05-29 20:50:30,No
23,1023,10.50,New York,Convenience Store,2024-05-29 07:45:20,No
24,1018,4000.00,Hawai,Jewelry,2024-05-29 18:30:00,Yes
25,1025,30.75,Colorado,Restaurants,2024-05-29 13:35:10,No
```

Column Descriptions:

TransactionID: Unique identifier for each transaction.
CardholderID: Unique identifier for each cardholder.
PurchaseAmount: Amount of money spent in the transaction.
Location: Location where the transaction occurred.
MerchantCategory: Category of the merchant (e.g., Groceries, Electronics, Restaurants).
TimeOfPurchase: Timestamp of when the purchase was made (in the format YYYY-MM-DD HH:MM).
Fraudulent: Indicates whether the transaction was identified as fraudulent (Yes, No).

Describe one anomaly detection technique you could employ to analyze credit card transaction data. Briefly explain the concept behind the technique.

Q.4) (20) A streaming service offers a vast library of movies and TV shows. They have collected user data on viewing habits, including genres watched, duration of viewing sessions, and time of day for watching. The streaming service wants to understand their customer base better and personalize their recommendations.

Here is a sample dataset in CSV format based on the problem definition. This dataset includes user data on viewing habits, such as genres watched, duration of viewing sessions, and time of day for watching:

UserID	Age	Gender	GenresWatched	Avg SessionDuration	Total WatchTime	Preferred WatchTime
1	25	Male	Action|Comedy	45	1200	Night
2	34	Female	Drama|Romance	60	1500	Evening
3	29	Male	Documentary|Science	30	800	Morning
4	42	Female	Action|Thriller	50	2000	Night
5	23	Male	Comedy|Drama	35	900	Afternoon
6	37	Female	Horror|Mystery	40	1000	Evening
7	31	Male	Action|Sci-Fi	55	1800	Night
8	28	Female	Romance|Comedy	45	1100	Afternoon
9	45	Male	Drama|Historical	60	2200	Evening
10	30	Female	Sci-Fi|Fantasy	35	950	Morning
11	26	Male	Comedy|Action	40	1050	Night
12	33	Female	Mystery|Thriller	50	1250	Evening

Column Descriptions:

UserID: Unique identifier for each user.

Age: Age of the user.

Gender: Gender of the user.

GenresWatched: Genres of content the user frequently watches (multiple genres separated by '|').

AvgSessionDuration: Average duration of each viewing session in minutes.

TotalWatchTime: Total watch time in minutes.

PreferredWatchTime: Preferred time of day for watching content (Morning, Afternoon, Evening, Night).

a) (10) Identify a suitable unsupervised learning algorithm for segmenting the users based on their viewing habits. Briefly explain why this approach is appropriate for this problem.

b) (10) Describe the key steps involved in segmenting the users using the chosen algorithm. Briefly explain what happens at each step.

Q.5) (20) A retail store has provided you with a sample of their daily sales data for the past week in a CSV file named "sales_data.csv". The data includes columns such as date, product category (category), product ID (product_id), quantity sold (quantity), and total sales amount (sales_amount).

Sample Data (sales_data.csv):

```
date,category,product_id,quantity,sales_amount
2024-05-20,Electronics, TV123, 2, 1200.00
2024-05-20,Clothing, Shirt456, 1, 35.99
2024-05-21,Appliances, Fridge789, 1, 899.99
2024-05-21,Grocery, Milk123, 3, 5.99
2024-05-22,Electronics, Monitor007, 1, 250.00
2024-05-22,Clothing, Pants987, 2, 79.98
2024-05-23,Appliances, Washer345, 2, 599.98
2024-05-23,Grocery, Eggs567, 1, 2.99
```

a) (10) Calculate the total sales per product category for the past week

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")
sales_df["date"] = pd.to_datetime(sales_df["date"])
today = pd.to_datetime('today')
last_week_start = today - pd.Timedelta(days=7)
print(sales_df[ ____________ ]. ______ ( ______ )[ ________ ].sum())
```

b) (10) Identify the top 5 products (by product ID) that generated the most revenue during the past week.

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")
sales_df["date"] = pd.to_datetime(sales_df["date"])
today = pd.to_datetime('today')
last_week_start = today - pd.Timedelta(days=7)
total_revenue_per_product = ______________________ .sum()

sorted_revenue = total_revenue_per_product.sort_values(ascending=False)

top_5_products = sorted_revenue. ________
print(top_5_products)
```